

On Mathematical Inference

and its application to various domains

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*“I got bored one day and I put everything on a bagel... **everything:***

All my hopes and dreams, my old report cards, every breed of dog, every last personal ad on Craigslist, sesame, poppy seed, salt... and it collapsed in on itself. 'Cause you see, when you really put everything on a bagel, it becomes this: The truth.”

Jobu Topaki in Everything Everywhere All at Once

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PART I

Prelude: Andante **Thematic introduction**

*“Science, including philosophy, is
intrinsically polemical, and the
polemical cannot be separated from the
agonistic. In epochs in which great new
things arise, the agonistic factor
usually comes to the fore”*

Johan Huizinga in *Homo Ludens*

CHAPTER 1

Inspiration

When I was an undergraduate studying physics, the probabilistic side of statistical and quantum theory often felt like a maze of disconnected rules and interpretations. Concepts such as ‘subjective versus objective probability’ and the clash between Bayesian and frequentist viewpoints seemed to sit at odds with logic.

Everything changed when I discovered Edwin Thompson Jaynes’ book *Probability Theory* after my studies. The book, written by a fellow physicist that struggled with very similar problems, gave me a single, coherent picture: Probability can in fact be seen as extended logic, and the apparent contradictions in the literature dissolve once we treat inference as a systematic process. Jaynes’ ideas have shaped every subsequent step of my thinking, and his work remains a fixed point for my master’s thesis in artificial intelligence.

Jaynes’ book, however, stops abruptly, since he died in the process of making it. As with Gogol’s *Dead Souls* it made me curious what his finished work would have looked like: What parts would have been added, what parts would have been revised, etc. In this sense this thesis is a modest attempt to finish what Jaynes began.

In short, this thesis is both a genuine tribute to a thinker who reshaped my understanding of probability and a concrete step toward finishing the story he started.

CHAPTER 2

Introduction

Jaynes' work

As David Thomas and Andrew Hunt, co-writers of the Agile Manifesto and authors of the highly influential book *The Pragmatic Programmer* have put it: “A good idea is an orphan without effective communication. As developers, we have to communicate on many levels. We spend hours in meetings, listening and talking. We work with end users, trying to understand their needs. We write code, which communicates our intentions to a machine and documents our thinking for future generations of developers.” In fact, the same holds true for practically anyone with a good idea – of which Jaynes was certainly not short – except that aforementioned ‘code’ should be replaced by any kind of medium that transfers information.

Jaynes’ book *Probability Theory* – and his work in general – is experienced as an extremely polemic work. In the light of Huizinga’s view of science (see the opening of this movement), it can be concluded that something paramount was at stake for Jaynes. For Jaynes it was essential that the then current – nowadays still largely dominant – praxis in several scientific and philosophical domains would be completely overturned, since he deemed them incorrect; not an amiable message to the scientific community. Due to this highly polemic tone, and the fact that Jaynes wasn’t always able to deliver a viable, structured alternative, his work has, however, gone largely unnoticed.

The book concerns itself mainly with the process of inducing new knowledge from available information. As Jaynes rightly noticed [REFERENCE], this is – or at any rate *should be* – the process in which the scientific process takes place. Jaynes presented his view vehemently as an anti-movement, where he delivered critiques on both the mathematical praxis within scientific research, as well as on the mathematical theory that in many areas moves general knowledge forward. For the theory of statistics specifically, he was able to deliver a convincing alternative within the book [REFERENCE], but many other critiques were left unsolved, nor did Jaynes manage to create an overarching view on how all these problems were linked together.

Current work

In the present work, I address the core critiques Jaynes delivered in a more structured and – hopefully – more friendly and manageable manner. Furthermore, I aim to build upon the main insights taken from his work, which

2. Introduction

are too important to be lost for future generations. Both his critiques and insights are embedded in a broader, self-consistent structure I took the liberty calling 'mathematical induction' – referring here to an iterative, principle-driven way of building knowledge, not to the elementary proof technique.

The contribution of this dissertation can be divided in four parts:

1. It reorganises Jaynes' critiques into a unified methodology
2. It supplies concrete case studies that illustrate how the methodology avoids the pitfalls Jaynes identified
3. It investigates why current mathematical and scientific praxis was ineffective finding these solutions
4. It proposes changes to this praxis, so that future generations will be more effective in their researches

The core critiques fall into three overarching themes: Methodology, discreteness and probability. These critiques are examined in detail throughout the dissertation:

Methodology

Methodology vs. gamesmanship

Jaynes warned against 'mathematical gamesmanship' – the temptation to manipulate formulas for short-term gain – and advocated a systematic, principle-based method. Mathematical induction follows a structured methodology, which is described in chapter 5. Throughout this work, this method is applied repeatedly.

Conceptual understanding vs. over-definition/showcasing mathematical skills

He argued that getting lost in formal definitions often obscures the real, underlying problem. This phenomenon can sometimes be seen as a variation of previous problem, where gamesmanship or showing off mathematical skills is greatly rewarded compared to conceptual understanding. Mathematical induction introduces definitions only when they serve the logical flow, and keeps the focus on the underlying concepts, as explicated in chapter 5. This work in particular, often keeps the major part of the mathematical machinery out of the main text and moves them to the appendix [REFERENCE], where the interested reader can follow the details.

Discreteness

Discrete vs. continuous

Jaynes contrasted treating infinity as an abstract entity with the safer practice of beginning from a finite case and then taking a limit. Mathematical induction doesn't allow for such unsafe practices, but builds and extends on finite models. The first case study that showcases my methodology involves the improper use of limits.

Probability

Bayesian/subjective/epistemic probability vs. frequentist/objective/physical probability

The book juxtaposes Bayesian (subjective) and Frequentist (objective) viewpoints, as well as epistemic versus physical probability. He advocated for a logical – yet subjective – interpretation of probability. Mathematical induction considers probability as a quantification of the degree of personal belief that is updated by evidence. What this means for these opposing views is discussed at great length in [REFERENCE].

Unknown vs. random

This can be seen as a variation of previous point, where information is considered unknown in a ‘subjective’ viewpoint, but ‘random’ in an absolute viewpoint. Jaynes did not believe in random variables. Mathematical induction doesn’t work with random variables. This point is discussed shortly in [REFERENCE].

Bayesian/subjective/epistemic probability vs. quantum probability

Jaynes denounced the modern interpretation of probability in quantum physics. He spotted a physical, inconsistent interpretation in its regular practice. As phrased before, mathematical induction considers probability as a quantification of the degree of personal belief that is updated by evidence. What this means for quantum physics is discussed at great length in [REFERENCE].

Therefore, by adopting a disciplined method, I reconcile the tensions Jaynes highlighted and extend the logical foundation to contemporary scientific practice. My hopes are that this work will contribute to a better understanding and wider appreciation of Jaynes’ work at large.

CHAPTER 3

Outline

Current monograph has the following structure:

Prelude – Andante: Thematic introduction

The opening sets the philosophical backdrop, explains why the thesis is framed as a symphony and notes the other idiosyncrasies dispersed throught the text. It establishes the “key signature” that recurs throughout the work.

First movement – Allegro: A structural method for doing mathematical induction

- Defines the core framework: Mathematical induction.
- Contrasts it with traditional mathematical praxis and Jaynes’ probabilistic logic.
- Introduces the “finite-to-infinite” principle that underlies everything else.

Second movement – Andante: Logic and probability as a basis for mathematical induction

- Builds logical circuits and probabilistic reasoning from the ground up.
- Covers logical gates, belief-updating circuits, entropy, odds, evidence, testing, and the relationship between distributions.
- Supplies the practical machinery that later movements will scale up.

Third movement – Scherzo: Exempla of mathematical induction

- Statistical “matter” – correcting the foundations of (statistical) thermodynamics.
- Quantum probability – clarifying epistemological misuse.
- Economic models – structural application.
- Medical examples – structural discovery.

3. Outline

Fourth movement – Vivace: Mathematical induction from circuits to networks

- Lifts the inductive machinery from low-level circuits to graph-theoretic networks and multi-layered learning systems.
- Discusses autonomous structure detection, continual-learning architectures, and rescaling techniques.
- Bridges the gap between basic components and full-blown AI systems.

Finale – Presto: Pragmatic suggestions

- Translates theory into actionable guidance for AI practitioners.
- Presents ultra-fast inference with a traveling-salesman at lightspeed.
- Investigates the question of what happens during physical experiments.
- Offers a concise scientific manifesto outlining a research agenda.

Appendix, acknowledgements & sources

- Appendix – provides detailed proofs, source code for the inductive algorithms, and a glossary of current and alternative terminology.
- Acknowledgements – credits collaborators and intellectual ancestors.
- Sources – lists papers, books, images, and inspirational texts for further reading.

Reading strategy

- Proceed sequentially through the movements; each builds directly on the previous one, much like themes in a symphony.
- Consult the Appendix whenever you need formal proofs, runnable code, or precise definitions.

By the end of this text the author hopes you, the reader, will be able to:

1. Articulate a unified inductive methodology.
2. Apply it to problems ranging from inference with relatively few variables to deep-learning network design.
3. Extend the framework to new domains, guided by the methodology and examples presented herein.

CHAPTER 4

Idiosyncrasies

This work adopts several personal choices, including – but not limited to – the following:

Inconventional structure

The present work is deliberately shaped like a symphony. The metaphor is taken from Johan Huizinga’s *Homo Ludens* and from the observation that the playful spirit that once played a central part in science has, in the last century, been mostly lost, since its practice became overly institutionalised. Each “movement” of the thesis therefore follows a musical logic rather than a conventional layout.

Stylistic choices

- Minimal acronyms – The field of science has recently become saturated with abbreviations; to keep the text readable I have avoided them almost entirely.
- Lower-case preference – Whenever grammar permits, nouns are left in lower case. This reduces unnecessary capitalization and gives the prose a smoother visual rhythm.
- Therefore: No terms like ‘MILE’ (Mathematical Induction as Logical Extension), but simply “mathematical induction”.

Alternative naming

- Strictly speaking the term ‘mathematical induction’ for the proof technique is a misnomer; ‘mathematical deduction’ would, in fact, have been much more apt. Therefore, I have decided to call it such, which ‘frees up’ the term ‘mathematical induction’ for my own methodology, which I consider to be a quite accurate description for it.
- Although discussed rather briefly, the term ‘thermodynamics’ is actually a gross misnomer for this domain, since no time-dependencies or dynamics are involved in it. Therefore, ‘thermostatics’ should have been the adopted term, and I will name the field as such.

4. Idiosyncrasies

- When speaking about the various Linux flavours I, like many others, use the informal term “Linux Distro’s”, rather than the more formal “Linux distributions”. In the spirit of consistency, I have therefore decided that probabilistic densities are also referred to as “distro’s” throughout the manuscript.
- Current and alternative terms can be found in the glossary [REFERENCE].

Audience considerations

The thesis is crafted with two very different audiences in mind. On the one hand, people in the ‘hard’ sciences, that will find the formal development of the inductive method familiar and rewarding; on the other, people trained in the humanities that will be drawn to the conceptual reflections that underpin the work. Because the text strives to satisfy both communities, it can sometimes feel richly detailed for a mathematically inclined reader while appearing somewhat sparse for someone rooted in the humanities. Nonetheless, the intention is that each audience will be patient with the hybrid style and discover value in the cross-disciplinary approach.

Sources

As is customary in the humanities, books play a central role in shaping the conceptual background of this work. Extended readings allow me to trace the evolution of ideas that papers, constrained by length, often leave unmentioned. The thesis therefore leans heavily on monographs and long texts to capture the “trivialities” – the small questions that reveal the larger purpose of a researcher’s inquiry.

Collaboration with AI

This text is authored by me, but it was written in collaboration with Lumo, the open-source and privacy-focused AI from Proton. Lumo helped me to search for references and polish my prose. The result embodies a new mode of scientific collaboration in which the researcher retains final intellectual and editing responsibility for the work, but valuable additions are made by AI.

Placement of proofs

For the sake of narrative flow, the majority of formal proofs have been relocated to the appendix. This allows the main text to concentrate on the development of ideas without interruption, while still providing complete, verifiable demonstrations for readers who wish to check the conclusions.

These idiosyncrasies are not decorative quirks; they are intentional decisions that support the overarching goal of presenting a cohesive, playful, and accessible exploration of mathematical induction as a logical extension of scientific reasoning.

PART II

First movement: Allegro **A structural method for doing** **mathematical induction**

*“Let mathematicians and astronomers
save themselves if they can with
infinitely disappearing minute
magnitudes, but in life itself this does
not help a man to obtain his
examination papers, and much less to
explain spirit”*

Vigilius Haufnensius in *Concept of Anxiety*

CHAPTER 5

Manifesto for mathematics

This work is heavily indebted to the ‘manifesto for agile software development’ [REFERENCE]. Although aimed at producing useable and reusable software in production environments, its practical wisdom is applicable to domains outside of its origins – including mathematics. Since mathematical theories are only useable in as far as they are reusable, like software, the following contains a comparable manifesto for the development of mathematical theories.

Pragmatic manifesto for the development of mathematical theories

I am uncovering better ways of developing mathematical theories by doing it and helping others do it. Through this work I have come to value:

Algorithms over abstractions

Narrative over formulae

Explicit over implicit

Simple over complex

That is, while there is value in the items on the right, I value the items on the left more.

Consequences

This manifesto handles the commentary Jaynes delivered on mathematical praxis at large. It focuses on reproducibility rather than showmanship; and on motivation rather than extensive formulae. This manifesto actively promotes the conceptual understanding behind advancing certain theories by making it practical, insightful and explicit.

Critics may confuse ‘simple’ with ‘simplistic’, which is not the same. Furthermore, some critics may confuse mathematics with theorems that come about in a flash of insight as a work of lonely genius. Make no mistake: Mathematics is a concerted effort – one that benefits from good communication and a structured methodology – and it should unambiguously be treated as such.

The remainder of this thesis showcases the results of this manifesto. The finale [REFERENCE] discusses possible mechanisms that previously hindered such a practical approach and proposes solutions for these obstructions.

CHAPTER 6

To infinity and back again

As a first case study for how mathematical praxis got off track by focusing too heavily on abstractions, formulae, proofs and showmanship, rather than algorithms, motivation, narrative and patience, I will briefly detail the proof of Cantor. Jaynes made it abundantly clear that he did not prefer the methodology of Cantor and his followers, but didn't clarify what was exactly wrong with it. In the following I will use the mathematical manifesto to bridge this omission.

Note that in the following text the proofs are provided in the text rather than in the appendix [REFERENCE], since the details of the proofs are needed in order to refine them.

Different kinds of infinity

There is a famous proof of Cantor [REFERENCE], that although the natural numbers ($\mathbb{N}; 1, 2, 3, \dots$), fractions ($\mathbb{Q}; \frac{2}{3}, \frac{5}{17}, \frac{7}{1}, \dots$) and the real numbers ($\mathbb{R}; \sqrt{2}, \pi, e, \dots$) are all 'infinite', their 'level' of infinity differs; that is: $\mathbb{R}$ is deemed 'more infinite' than $\mathbb{N}$ and $\mathbb{Q}$ – which are deemed the same 'level' of infinity.

The proof follows the following logic:

Step 1: Assume the natural numbers, then fractions can be derived if you introduce multiplications and divisions. Yet, more numbers can be derived if you allow for exponentiation – a noncomplete definition of real numbers.

Step 2: There are real numbers that cannot be expressed as fractions. This is proven by assuming the opposite, so that, for example:

$$\sqrt{2} = \frac{n_1}{n_2}$$

where n_1 and n_2 are natural numbers without common factors, that is, are coprime. By squaring, the result becomes:

$$2 = \frac{n_1^2}{n_2^2} \implies n_1^2 = 2n_2^2$$

which means that n_1^2 is even (can be divided by two). However, if a square of an integer is even, the number *itself* must also be even, since, suppose that n_1^2 is odd, you can write $n_1 = 2n_3 + 1$ for n_3 any natural number; so that:

$$n_1^2 = (2n_3 + 1)^2 = 4n_3^2 + 4n_3 + 1 = 2(2n_3^2 + 2n_3) + 1$$

6. To infinity and back again

an odd number, a contradiction. However, if n_1 is even, you can likewise write $p = 2n_3$ for n_3 a natural number. If I substitute this back into the equation above:

$$(2n_3)^2 = 2n_2^2 \implies 4n_3^2 = 2n_2^2 \implies n_2^2 = 2n_3^2$$

which means that n_2^2 and, therefore, n_2 itself are also even. This contradicts the condition that n_1 and n_2 are natural numbers without common factors. Naturally, the proof can be repeated for any square root number, say $\sqrt{3}$.

Step 3: Proof that there is a bijection between $\mathbb{N}$ and $\mathbb{Q}$, that is: Have the same cardinality. This can be done by representing the fractions such:

	$n_2 = 1$	$n_2 = 2$	$n_2 = 3$	$n_2 = 4$	$n_2 = 5$	...
$n_1 = 1$	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	...
$n_1 = 2$	$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	...
$n_1 = 3$	$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	...
$n_1 = 4$	$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	...
$n_1 = 5$	$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Now, I can walk through this table in an anti-diagonal fashion; circling outward, starting with $\frac{1}{1}$. In other words: $n_1 + n_2 = n_3$ (with $n_3 \geq 2$).

$$\begin{aligned}
 D_2 &: (1, 1) \\
 D_3 &: (1, 2), (2, 1) \\
 D_4 &: (1, 3), (2, 2), (3, 1) \\
 D_5 &: (1, 4), (2, 3), (3, 2), (4, 1) \\
 &\vdots
 \end{aligned}$$

This is almost complete, except that zero and the negative numbers still need to be included. This can be done by starting the sequence with 0, then continue with D_2 , then with $-D_2$ (that is, all numbers in D_2 , but negative), then with D_3 , then $-D_3$, etc. Important is to skip fractions that are not fully reduced, that is, still share a factor, e.g.: $\frac{2}{2}$.

This is a bijection, since:

- **Injectivity.** All fractions that are not fully reduced are omitted, leaving only unique fractions in the process.
- **Surjectivity.** At the first step 0 is included; for every other reduced fraction $\frac{n_1}{n_2}$, the fraction occurs in row n_1 and column n_2 , which means it occurs in the sequence when D_{n_3} appears such that $n_1 + n_2 = n_3$; likewise for every negative counterpart.

This process therefore establishes that:

$$|\mathbb{N}| = |\mathbb{Q}|$$

Step 4: Proof that $\mathbb{N}$ and $\mathbb{R}$ cannot be mapped by a similar procedure. This can be done by first assuming such a map, using:

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}d_{14}\dots \\ r_2 &= 0.d_{21}d_{22}d_{23}d_{24}\dots \\ r_3 &= 0.d_{31}d_{32}d_{33}d_{34}\dots \\ &\vdots \end{aligned}$$

where each decimal is written in a non-terminating fashion, for example $0.4999\dots$ is written as $0.5000\dots$

I now define a new decimal:

$$r^* = 0.d_1d_2d_3\dots, \quad d_n = \begin{cases} 5 & \text{if } d_{nn} \neq 5, \\ 6 & \text{if } d_{nn} = 5. \end{cases}$$

I can clearly see that, by construction, this decimal is not part of the, supposedly bijective, sequence of r_n 's. It is, however, a valid decimal, contradicting the original claim that such a sequence does, in fact, exist.

All the previous steps therefore establish that:

$$|\mathbb{R}| > |\mathbb{N}|$$

and since $|\mathbb{Q}| = |\mathbb{N}|$ it, likewise, follows that:

$$|\mathbb{R}| > |\mathbb{Q}|$$

Limits

Previous proofs use limits extensively, albeit implicitly. Let me build up the definition of a limit and revise previous proofs using that definition.

First, before we can define limits, we assume, identical to step 1 and together with Felix Klein, that the natural numbers $(\mathbb{N}; 0, 1, 2, \dots)$, are ‘god-given’ – a ‘machina ex deo’ if you will. I include 0 in $\mathbb{N}$ here, which is not necessary for the following, but a matter of taste. These are *constants*, the rest will be derived from constants (and are only constants themselves in so far they are equal to a value in $\mathbb{N}$, take $\frac{2}{2}$ for example). For example: Fractions can be derived by dividing two constants; Real numbers are then generally derived from fractions using Dedekind cuts.

A limit, $\lim_n$, is a process that forms a countable sequence. The underlying process itself is obscured – executed under the hood – but at every step n progresses. As $\mathbb{N}$ has no maximum value – you can always add one more – it means that limits can also ‘go to infinity’. The following is very important, as it is often left unspecified: Such a limit is always *ad infinitum*, meaning: Forever progressing, never terminated, a ‘moving target’.

Revising step 2

It can be proven, see [REFERENCE] that all real numbers can be expressed as a decimal expansion, such as:

$$1 = 1.000\dots$$

6. To infinity and back again

$$\frac{2}{3} = 0.6$$

$$\sqrt{2} = 1.4142\dots$$

These expansions are, in fact limits, where the dots, $\dots$, obscure a limit to infinity, $\lim_{n \rightarrow \infty}$. In the case of $\sqrt{2}$, for example, the limit obscures the process in which $\sqrt{2}^2$ is the constant 2. I can then write, in improvised notation:

$$\sqrt{2}(N) = \lim_{n \rightarrow N} 1.4142(n) : (\lim_{n \rightarrow \infty} 1.4142(n))^2 = 2$$

so that:

$$\sqrt{2} = \lim_{n \rightarrow \infty} 1.4142(n)$$

However, the first expression can alternatively be written as:

$$\sqrt{2}(N) = \lim_{n \rightarrow N} 1.4142(n) = \lim_{n \rightarrow N} \frac{14142(n)}{10^n} = \frac{n_1}{n_2}$$

so that it follows that in every step of the process, $\sqrt{2}$ can be written as a fraction. This means, since $\sqrt{2}$ can in every step of the process be written as a fraction, it can, even in the infinite limit, be written as a fraction. In fact, if you accept that every real number can be written as a decimal, it follows that every real number can in fact be written as a fraction – albeit an infinite one, against which there is in principal no objection.

This poses the question what part in step 2 of the previous chapter was invalid. Let's alternatively write:

$$\sqrt{2} = \lim_{n \rightarrow \infty} 1.4142(n) = \frac{\lim_{n \rightarrow \infty} 14142(n)}{\lim_{n \rightarrow \infty} 10^n} = \frac{n_1}{n_2}$$

this with the understanding that these limits are, in fact, linked to each other (every step in either one process, implies a step in the other process). This, in order to be able to replicate the steps of the proof. Let's continue:

$$2 = \frac{n_1^2}{n_2^2} \implies n_1^2 = 2n_2^2$$

which would, again, imply that n_1 is even. *However*, it is impossible to write $2n_3 = n_1$ since n_1 is an infinite process – a moving target – in which the underlying process in no way guarantees that this can be done. With this I mean that it is perfectly fine to *square* n_1 , but ‘taking out a factor’, is not guaranteed to work. Therefore, this operation is undefined and the proof can't proceed.

In fact, the classic proof only ‘makes sense’ if you allow for finite fractions. However, comparing infinitely extending real numbers only with finite fractions seems useful nor fair. If you compare infinitely extending real numbers with infinite fractions, it follows there is no obvious difference between fractions and real numbers.

Revising step 4

This poses the question what part of Cantor's second algorithm is invalid. Let's inspect:

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}d_{14}\dots \\ r_2 &= 0.d_{21}d_{22}d_{23}d_{24}\dots \\ r_3 &= 0.d_{31}d_{32}d_{33}d_{34}\dots \\ &\vdots \end{aligned}$$

The dots, $\dots$, are *actually* limits to infinity, although nicely covered up. That means that the 'trick':

$$r^* = 0.c_1c_2c_3\dots, \quad c_n = \begin{cases} 5 & \text{if } d_{nn} \neq 5, \\ 6 & \text{if } d_{nn} = 5. \end{cases}$$

is, in fact, invalid, since for any step in the process, if I define any such number r^* not yet in the sequence for r_n , I simply add it at the end; that is what it means to have a process *ad infinitum*.

Cantor's algorithm only 'makes sense' if you allow for infinite yet 'static' limits, that is: A sequence that is infinite, but 'fixed'. The latter option runs counter to every conceivable definition of having an infinite process – even the process of extending the natural numbers, $\mathbb{N}$, to infinity can't be understood in this fashion.

This particular way of treating infinite limits – as 'infinite', yet static – is something that Jaynes fiercely opposes¹, without making explicit what is *exactly* wrong with it, or proposing a viable alternative; Although he sometimes seems to allure to an extreme form of *finitism*, where he plainly denounces infinite processes altogether.

The problem Jaynes describes is genuine: Treating infinite limits as static holds the risk of treating such limits 'as a constant', disregarding the fact that such a limit always represents a non-terminating process. This, in turn, allows for risks such as acting on these limits with invalid operations, like: 'taking out a factor', creating subsequences consisting of terms from infinite sequences, and more. The alternative is treating infinite limits as a non-terminating process, that is: 'Dynamic'; Completely in line with Jaynes' general advice of starting with a finite case to check whether an operation is allowed, and then only *after* extending the process to infinity.

Alternative algorithm to Cantor's second algorithm

This leaves open the question whether there is in fact a process to count real numbers. Given that all real numbers can be expressed as decimals, I can use

¹He describes it, among other terms, as 'Bourbakiism', after the Bourbaki collective that frequently used this practice.

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the following algorithm:

$$\begin{array}{rcl}
 & & d_0 d_1 d_2 d_3 \dots \\
 n_0 & 0 & . \ 0000 \dots \\
 n_1 & 1 & . \ 1111 \dots \\
 n_2 & 2 & . \ 2222 \dots \\
 n_3 & 3 & . \ 3333 \dots \\
 n_4 & 4 & . \ 4444 \dots \\
 n_5 & 5 & . \ 5555 \dots \\
 n_6 & 6 & . \ 6666 \dots \\
 n_7 & 7 & . \ 7777 \dots \\
 n_8 & 8 & . \ 8888 \dots \\
 n_9 & 9 & . \ 9999 \dots \\
 n_{10} & 10 & . \\
 n_{11} & 11 & . \\
 n_{12} & 12 & . \\
 & \vdots &
 \end{array}$$

where every step combines all natural numbers on the outerleft up to a certain index, with all decimals up to the same index, alternating positive and negative numbers (except for 0). Thus: I start with n_0 and combine it with all numbers in d_0 , that is: $0.0; 0.1; -0.1; 0.2; \dots 0.9; -0.9$. Afterwards, I progress with 1, so I need to combine n_0 and n_1 with all decimals $d_0 d_1$, that is: $0.00; 0.01; -0.01; 0.02; \dots -1.98; 1.99; -1.99$.

This process is obviously surjective, since all decimal numbers will occur in this sequence. It is not injective (since some numbers occur multiple times, like 0.0 and 0.000000). However, this is no objection, since the algorithm implies that $|\mathbb{R}| \subseteq |\mathbb{N}|$. However, by definition ($\mathbb{R}$ as an extension of $\mathbb{N}$), you also have $|\mathbb{N}| \subseteq |\mathbb{R}|$, so that $|\mathbb{N}| = |\mathbb{R}|$. Since it was already established that $|\mathbb{N}| = |\mathbb{Q}|$ it follows immediately that $|\mathbb{Q}| = |\mathbb{R}|$, and, from the fact that all decimals can be written as (potentially infinite) fractions, it follows that, in fact, they are completely equal: $\mathbb{Q} = \mathbb{R}$.

Consequences

Previous chapter is a short case study on how the critiques that Jaynes delivers on mathematical praxis are not void, especially on general methodology and on using continuous spaces without caution – rather than discrete ones that are later extended. The consequences of mathematical gamesmanship and showcasing your ability to manipulate entities – without truly thinking what these entities or these manipulations *mean* – can easily lead to a mathematical ‘show-off’.

The mathematical manifesto defined in the previous chapter 5, promotes a different kind of practice. By focusing on algorithms over abstractions, this chapter was able to explicitly deliver a method for mapping two hitherto ‘unmappable’ spaces – simply by sitting down and doing so. In choosing narrative over formulae, a simple (but not simplistic, I stress) objection to two previously accepted theorems could easily be put forth. By making explicit what my definition of a limit is, and what hidden assumptions the other scientists must have had, I am able to highlight the exact points in which I think they went astray. For critics that would say: “This is rather simple” (as if that were something bad), I would reply: “Correct, but the argument went unnoticed for over a century – on who, then, is the joke?”. Therefore, this critique is more properly addressed to the overcomplication that took place, which is a process that apparently went unchecked for over a century.

The consequences are ample: The Pythagoreans were right after all, albeit in a manner I don’t think they envisioned as the definition of beauty. The construction of $\mathbb{R}$ with Dedekind cuts or otherwise becomes void, since there is no clear basis to assume that there are real numbers that cannot be expressed as fractions. As such, $\mathbb{R}$ completely collapses to $\mathbb{Q}$. This generally holds no severe consequences for real analysis, since, rather than assuming that ‘ $\mathbb{Q}$ is not dense enough, but $\mathbb{R}$ is’, previous chapter only goes to show that $\mathbb{Q}$ is, in fact, much more dense than formerly thought.

[TODO: MORE CONSEQUENCES]

PART III

Second movement: Andante Logic and probability as a basis for mathematical induction

“Scientists have calculated that the chance of anything so patently absurd actually existing are millions to one.

But magicians have calculated that million-to-one chances crop up nine times out of ten”

Terry Pratchett in *Mort*

PART IV

Third movement: Scherzo **Exempla of mathematical induction**

“Als je dus een vakman bent en je me, zoals ik net zei, bedrogen hebt met je verzekering een Homeros-demonstratie te geven, lever je me een streek. Als je geen vakman bent, maar je door goddelijke gave van Homeros bezeten raakt en je in onwetendheid veel mooie dingen over de dichter zegt – wat zoals ik beweerde het geval is met jou – dan lever je me geen streek. Je hoeft dus maar te kiezen hoe je door ons aangemerkt wil worden, als een man die streken levert of een man die door de goden wordt beheerst”

Socrates in *Ion*

PART V

Fourth movement: Vivace Mathematical induction from circuits to networks

“The important thing is life — life and nothing else! What is any ‘discovery’ whatever compared with the incessant, eternal discovery of life?”

Hippolyte in *Idiot*

PART VI

Finale: Presto **Pragmatic suggestions**

*“Of course you can try to calculate
how much fruit you will attain from
your labors, but it won’t be accurate,
and there’s no need to do that. Just
plant the seeds and eventually you will
reap the harvest”*

Chan Master Sheng Yen in *Tea Words*

Appendices

“Contrarily, it is characteristic of a truly wise man that, being just a mortal, he doesn’t want to transgress any boundaries with his wisdom and together with humanity either turns a blind eye, or, out of courtesy, wanders along”

Folly in *The Praise of Folly*

PART VII

Acknowledgements and sources

“Teleutias voer werkelijk onder de gelukkigste omstandigheden terug naar huis. Toen hij naar zee afdaalde om aan de thuisreis te beginnen, was er geen soldaat die hem niet de hand drukte. De één zette hem een krans in het haar, de ander deed hem een hoofdband om en degenen die te laat kwamen gooiden, toen hij al uitvoer, toch nog kransen in zee en wensten hem alle goeds.

Ik beseef wel dat ik hiermee geen grote uitgave, geen gevaar, geen strategie die het vermelden waard is beschrijf, maar bij Zeus, het is toch voor een man naar mijn idee belangrijk om te bedenken met welk gedrag Teleutias eigenlijk zo’n reactie bij zijn ondergeschikten teweegbracht, want dat is, bij hoeveel geld en gevaren ook, de meest vermeldenswaardige prestatie van een man”

Xenofon in *Griekse Oorlogen*

